WORKED SOLUTIONS

Full-Length Practice Paper | Mathematics Class 9 / SSC-I

All MCQs and both alternatives in Sections B and C

These solutions are for the accompanying independent practice paper. Suggested subpart marks are practice allocations, not an official FBISE marking scheme.

SECTION A | Correct answers and reasons

No.	Option	Reason
1	C	$(1/16)^{-3/4} = 16^{3/4} = 2^3 = 8$.
2	A	$0.00072 = 7.2 \times 10^{-4}$; the leading number lies between 1 and 10.
3	B	The first coordinates are $1, 2, 4$.
4	C	$\gcd(12, 18) = 6$; take the smaller power of each variable: $6xy$.
5	D	Divide by -2 and reverse the inequality: $x \geq -3$.
6	A	For an acute angle, $\cos\theta = \sqrt{1 - 9/25} = 4/5$.
7	B	Reverse bearing $= 075^\circ + 180^\circ = 255^\circ$.
8	C	Length ratio $= \sqrt{9} : \sqrt{25} = 3 : 5$.
9	D	$1 + \cot^2\theta = \operatorname{cosec}^2\theta$.
10	A	Altitudes meet at the orthocentre.
11	B	$n = 360/24 = 15$.
12	C	Points equidistant from two fixed points lie on the perpendicular bisector.
13	D	Ordered data: $3, 5, 7, 8, 12$; the middle value is 7.
14	A	Primes: $2, 3, 5, 7, 11, 13, 17, 19$. Probability $= 8/20 = 2/5$.
15	B	Relative frequency $= 18/60 = 0.30$.

Before checking written answers

A numerical answer alone is not a complete model solution. Check the formula or reasoning, substitutions, arithmetic, units, and the final statement. Equivalent correct methods are acceptable.

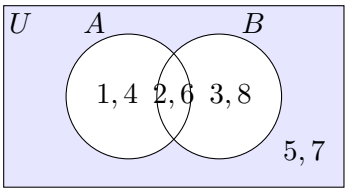
SECTION B | Q2(i)–Q2(iii)

Q2(i): Indices [4 marks]

$$\frac{27^{2/3} \times 16^{3/4}}{8^{1/3}} = \frac{(3^3)^{2/3}(2^4)^{3/4}}{(2^3)^{1/3}} = \frac{3^2 \times 2^3}{2} = \boxed{36}.$$

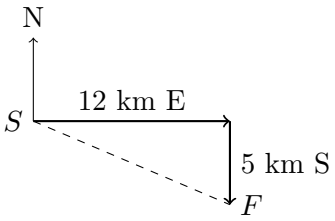
OR: Sets and Venn diagrams

$A \cup B = \{1, 2, 3, 4, 6, 8\}$, so $(A \cup B)' = \{5, 7\}$.
$A' = \{3, 5, 7, 8\}$ and $B' = \{1, 4, 5, 7\}$, so $A' \cap B' = \{5, 7\}$.
Hence $\boxed{(A \cup B)' = A' \cap B'}$.



The shaded outside region is both $(A \cup B)'$ and the region belonging to neither A nor B, i.e. $A' \cap B'$.

Q2(ii): Bearing [4 marks]



Angle south of east $= \tan^{-1}(5/12) = 22.62°$.
Bearing clockwise from north $= 90° + 22.62° = 112.62°$.
To the nearest degree, the three-figure bearing is $\boxed{113°}$.

OR: Cartesian product and relation

$A \times B = \{(1, 2), (1, 4), (3, 2), (3, 4), (5, 2), (5, 4)\}$.
Selecting pairs with $x > y$ gives $R = \{(3, 2), (5, 2), (5, 4)\}$.
$\boxed{\text{Domain}(R) = \{3, 5\}, \quad \text{Range}(R) = \{2, 4\}}$.

Q2(iii): Earthquake intensities [4 marks]

$$7.9 - 7.2 = \log_{10}(I_1/I_0) - \log_{10}(I_2/I_0) = \log_{10}(I_1/I_2).$$

Thus $I_1/I_2 = 10^{0.7} = \boxed{5.01}$ approximately. The first earthquake has about 5.01 times the intensity of the second.

OR: Line through an intersection

Adding $x + y = 5$ and $x - y = 1$ gives $2x = 6$, so $x = 3$, $y = 2$.
The parallel line has slope 2 because $2x - y + 4 = 0$ gives $y = 2x + 4$.
$y - 2 = 2(x - 3)$, hence $\boxed{y = 2x - 4}$.

SECTION B | Q2(iv)–Q2(vi)

Q2(iv): Compound inequality [4 marks]

Both inequalities must hold:

$$\begin{aligned} -3x+6 &\le 2x+1 \\ 5 &\le 5x \\ x &\ge 1 \end{aligned} \qquad \begin{aligned} 2x+1 &\le 4x+9 \\ -8 &\le 2x \\ x &\ge -4. \end{aligned}$$

Their intersection is $\boxed{x \ge 1}$, or $[1, \infty)$.



OR: Factorization by substitution

Let $t = x^2 + 3x + 2$. Then

$$\begin{aligned} (x^2+3x+1)(x^2+3x+3) - 3 &= (t-1)(t+1) - 3 \\ &= t^2 - 4 = (t-2)(t+2) \\ &= (x^2+3x)(x^2+3x+4) \\ &= \boxed{x(x+3)(x^2+3x+4)}. \end{aligned}$$

The remaining quadratic has discriminant $9 - 16 = -7 < 0$, so it has no further real linear factors.

Q2(v): Trigonometric identity [4 marks]

$$\begin{aligned} \text{LHS} &= \frac{(1+\sin\theta)+(1-\sin\theta)}{(1-\sin\theta)(1+\sin\theta)} \\ &= \frac{2}{1-\sin^2\theta} = \frac{2}{\cos^2\theta} = 2\sec^2\theta = \text{RHS}. \end{aligned}$$

Thus the identity holds wherever both sides are defined.

OR: Regular polygon

$n = 360^\circ/18^\circ = \boxed{20}$ sides.
Each interior angle $= 180^\circ - 18^\circ = \boxed{162^\circ}$.
Perimeter $= 20 \times 2.5 = \boxed{50 \text{ cm}}$.

Q2(vi): Slope and line equation [4 marks]

$$m = \frac{10-1}{4-(-2)} = \frac{9}{6} = \frac{3}{2}.$$

Using $A(-2,1)$, $y - 1 = \frac{3}{2}(x+2)$, hence $\boxed{y = \frac{3}{2}x + 4}$.

OR: Probability

There are $5 + 3 + 2 = 10$ equally likely balls.
(a) $P(\text{blue}) = \boxed{3/10}$.
(b) $P(\text{not green}) = (5+3)/10 = \boxed{4/5}$.
(c) $P(\text{red or green}) = (5+2)/10 = \boxed{7/10}$.
(d) $P(\text{red}) + P(\text{blue}) + P(\text{green}) = 5/10 + 3/10 + 2/10 = \boxed{1}$.

SECTION B | Q2(vii)–Q2(ix)

Q2(vii): Equidistant locus [4 marks]

The locus is the **perpendicular bisector of** PQ. For any point (x, y) on it, the squared distances are equal:

$$(x-2)^2 + (y-3)^2 = (x-8)^2 + (y-7)^2$$
$$x^2 - 4x + 4 + y^2 - 6y + 9 = x^2 - 16x + 64 + y^2 - 14y + 49$$
$$12x + 8y - 100 = 0.$$

Therefore $\boxed{3x + 2y - 25 = 0}$. Check: midpoint $(5, 5)$ satisfies the equation; its slope is $-3/2$, perpendicular to the slope $2/3$ of PQ.

OR: Perpendicular line

The given slope is -2. Thus the required slope is $-1/(-2) = 1/2$.
$y - 1 = \frac{1}{2}(x - 4)$, so $\boxed{y = \frac{1}{2}x - 1}$.

Q2(viii): Similar triangles [4 marks]

Area ratio $= 16 : 49$, so corresponding length ratio $= 4 : 7$.
If the larger corresponding side is s, then $8/s = 4/7$, giving $\boxed{s = 14 \text{ cm}}$.
Perimeters have the same ratio as corresponding lengths: $\boxed{4 : 7}$.

OR: Relative and expected frequencies

Observed tails $= 120 - 42 = 78$.
(a) Relative frequency of heads $= 42/120 = \boxed{0.35}$.
(b) Relative frequency of tails $= 78/120 = \boxed{0.65}$.
(c) For a fair coin, theoretical $P(H) = 1/2$. Expected heads $= 120(1/2) = \boxed{60}$. This is an expectation, not a claim that the observed count must be 60.

Q2(ix): HCF and LCM [4 marks]

$$x^2 - 9 = (x-3)(x+3), \qquad x^2 + x - 6 = (x+3)(x-2).$$

The common factor is $x + 3$, and the LCM uses every distinct factor once:

$$\boxed{\text{HCF} = x + 3, \qquad \text{LCM} = (x-3)(x+3)(x-2).}$$

OR: Weighted average

Runs in the three sessions are $10(8) = 80$, $25(6) = 150$, and $15(10) = 150$.
Total runs $= 80 + 150 + 150 = \boxed{380}$.
Average for all 50 overs $= 380/50 = \boxed{7.6 \text{ runs per over}}$.
The unweighted average $(8 + 6 + 10)/3$ is unsuitable because the sessions have different lengths.

SECTION C | Q3: Both alternatives

Perfect-square polynomial **[8 marks]**

A monic quartic square can be written as $(x^2+ax+b)^2$ (changing the overall sign of its quadratic square root does not change its square). Expanding,

$$(x^2+ax+b)^2 = x^4+2ax^3+(a^2+2b)x^2+2abx+b^2.$$

Compare with $x^4-8x^3+px^2+qx+9$:

$$2a=-8 \implies a=-4, \qquad b^2=9 \implies b=3 \text{ or } -3.$$

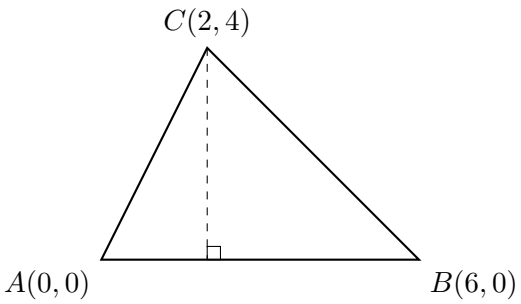
Case 1: $b=3$.
$p=a^2+2b=16+6=22$, $q=2ab=2(-4)(3)=-24$.
Thus $\boxed{(p,q)=(22,-24)}$ and the polynomial is $\boxed{(x^2-4x+3)^2}$.

Case 2: $b=-3$.
$p=16-6=10$, $q=2(-4)(-3)=24$.
Thus $\boxed{(p,q)=(10,24)}$ and the polynomial is $\boxed{(x^2-4x-3)^2}$.

Both pairs are required because the question asks for **all** possible pairs.

— OR —

Coordinate triangle **[8 marks]**



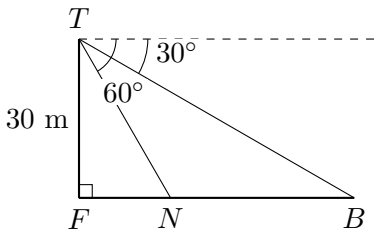
Slopes:

$$m_{AB}=\frac{0-0}{6-0}=0, \quad m_{AC}=\frac{4-0}{2-0}=2, \quad m_{BC}=\frac{4-0}{2-6}=-1.$$

At A, the angle above the positive horizontal is $\tan^{-1}(2)$, so $\angle A=\boxed{63.43^\circ}$ approximately.
At B, the ray BA points left and the ray BC points up-left. Its rise-to-horizontal-distance ratio is $4/(6-2)=1$. Therefore $\angle B=\tan^{-1}(1)=\boxed{45^\circ}$.
Then $\angle C=180^\circ-63.4349488^\circ-45^\circ=\boxed{71.57^\circ}$ approximately.
Check: $63.43^\circ+45^\circ+71.57^\circ=180^\circ$.

SECTION C | Q4: Both alternatives

Tower and two boats **[8 marks]**



N is the nearer boat, B the farther boat. Each angle of elevation equals the corresponding angle of depression because the horizontal lines are parallel.
Let $FN = d_n$ and $FB = d_f$. Then

$$\tan 60^\circ = \frac{30}{d_n} \implies d_n = \frac{30}{\sqrt{3}} = 10\sqrt{3}\ \text{m},$$

$$\tan 30^\circ = \frac{30}{d_f} \implies d_f = 30\sqrt{3}\ \text{m}.$$

Since both boats lie on the same side,

$$NB = d_f - d_n = 20\sqrt{3} = \boxed{34.64\ \text{m approximately}}.$$

If boats were on opposite sides, the distances would be added; that is not the situation here.

— OR —

Straight-line forms **[8 marks]**

(a) Slope-intercept form:

$$3x + 4y = 12 \implies \boxed{y = -\frac{3}{4}x + 3}.$$

(b) Two-intercepts form: Set $y = 0$ to obtain the x-intercept 4; set $x = 0$ to obtain the y-intercept 3. Dividing by 12,

$$\boxed{\frac{x}{4} + \frac{y}{3} = 1}.$$

(c) Symmetric form: Choose the point $(0, 3)$ and a direction angle θ with $\cos\theta = 4/5$, $\sin\theta = 3/5$; then $\tan\theta = -3/4$.

$$\boxed{\frac{x - 0}{4/5} = \frac{y - 3}{3/5}}.$$

This follows the form $(x - x_1)/\cos\theta = (y - y_1)/\sin\theta$. The equivalent direction-ratio form is $x/4 = (y - 3)/(-3)$. **(d) Normal form:** Divide by $\sqrt{3^2 + 4^2} = 5$:

$$\boxed{\frac{3}{5}x + \frac{4}{5}y = \frac{12}{5}}.$$

Thus $\cos\alpha = 3/5$, $\sin\alpha = 4/5$, and the perpendicular distance from the origin is $\boxed{p = 12/5 = 2.4}$ coordinate units.

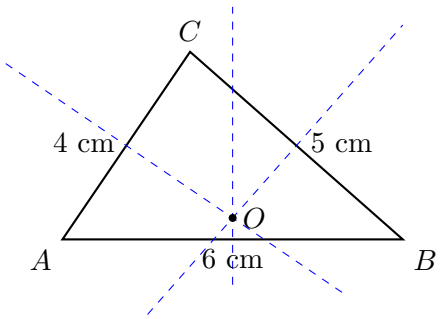
SECTION C | Q5: Both alternatives

Construction and perpendicular bisectors **[8 marks]**

Construction steps:

1. Draw the segment $AB = 6$ cm.
2. With centre A and radius 4 cm, draw an arc. With centre B and radius 5 cm, draw an arc meeting the first at C above AB.
3. Join AC and BC to complete $\triangle ABC$.
4. For AB, use equal-radius compass arcs of radius greater than half AB, centred at A and B, above and below the segment. Join the arc intersections to construct its perpendicular bisector.
5. Repeat the equal-radius arc procedure for BC and CA.
6. Label the common intersection O. It is the **circumcentre**; $OA = OB = OC$.

The diagram below illustrates the completed construction; students must make their own accurate ruler-and-compass drawing and leave construction arcs visible.



Concurrency follows because O on the bisectors of AB and AC gives $OA = OB$ and $OA = OC$, hence $OB = OC$. Therefore O also lies on the perpendicular bisector of BC.

— OR —

Grouped statistics **[8 marks]**

Class	Boundaries	f	Midpoint x	fx	Cumulative f
0–9	−0.5–9.5	4	4.5	18	4
10–19	9.5–19.5	6	14.5	87	10
20–29	19.5–29.5	10	24.5	245	20
30–39	29.5–39.5	8	34.5	276	28
40–49	39.5–49.5	2	44.5	89	30
Total		30		715	

Mean: $\bar{x} = \sum fx / \sum f = 715/30 = \boxed{23.83 \text{ units approximately}}$.

Median: $n/2 = 15$, so the median class is 20–29. Here $L = 19.5$, $c = 10$ (preceding cumulative frequency), $f = 10$, $h = 10$.

$$\text{Median} = L + \frac{n/2 - c}{f} h = 19.5 + \frac{15 - 10}{10}(10) = \boxed{24.5 \text{ units}}.$$

Mode: Modal class is 20–29; $f_m = 10$, $f_1 = 6$, $f_2 = 8$.

$$\text{Mode} = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} h = 19.5 + \frac{10 - 6}{20 - 6 - 8}(10) = \boxed{26.17 \text{ units approximately}}.$$